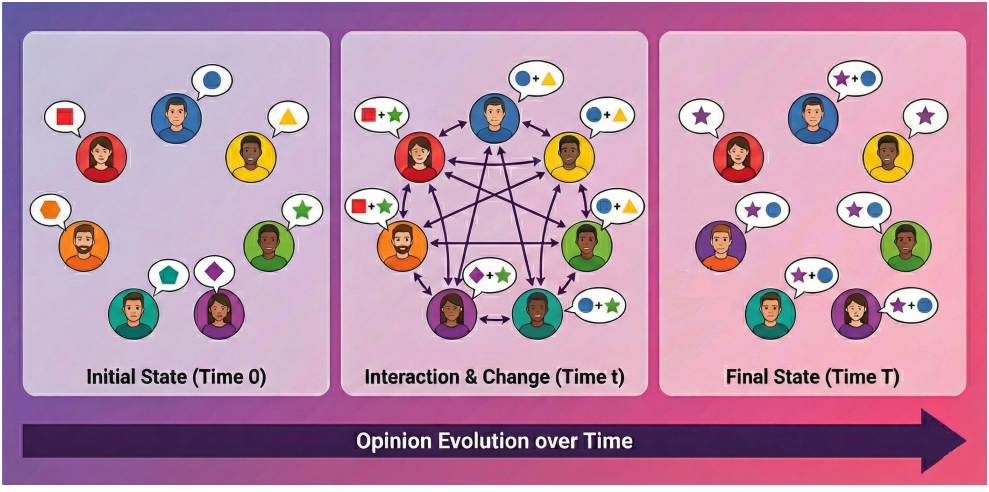
Optimal Input Allocation in DeGroot Model

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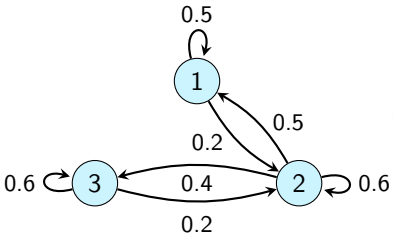
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Opinion Dynamics



DeGroot Model

Graph



Interaction Matrix (W)

From j

To i

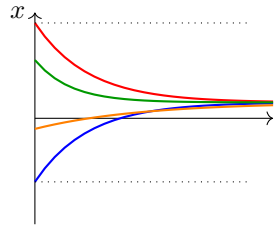
$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \mathbf{0.5} & \mathbf{0.5} & 0 \\ \mathbf{0.2} & \mathbf{0.6} & \mathbf{0.2} \\ 0 & \mathbf{0.4} & \mathbf{0.6} \end{bmatrix} \end{array}$$

Row Stochastic:

$$\sum_j W_{ij} = 1$$

Dynamics

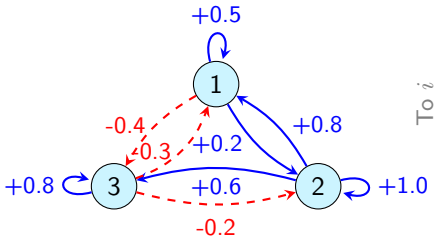
$$x(k+1) = Wx(k)$$



$$x^* = \mathbb{1}c^\top x(0)$$

Signed Opinion Dynamics

Signed Graph



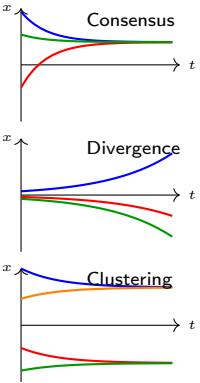
Interaction Matrix (W)

$$W = \begin{array}{c} \\ \text{To } i \\ \\ \\ \end{array} \begin{array}{c} \text{From } j \\ \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \mathbf{0.5} & \mathbf{0.8} & \mathbf{-0.3} \\ \mathbf{0.2} & \mathbf{1.0} & \mathbf{-0.2} \\ \mathbf{-0.4} & \mathbf{0.6} & \mathbf{0.8} \end{bmatrix} \end{array} \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}$$

Row Sum = 1:
$\sum_j w_{ij} = 1$

Dynamics Outcomes

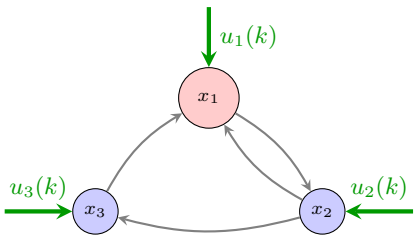
$$x(k+1) = Wx(k)$$



Problem Formulation

$$x(k+1) = Wx(k) + u(k)$$

Network



Cost

Volume: $u_{i,k}$ (Input)

Price: $p_i = \left(c_i / \min(c)\right)^{\gamma}, \quad \gamma > 0$

Total Cost $= p_i \times |u_{i,k}|$

The condition $\gamma > 1$ makes highly influential agents exponentially more expensive to influence.

Problem Formulation: The Objective Function

1. Network Convergence

The dynamics

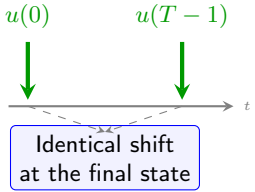
$$x(k+1) = Wx(k) + u(k)$$

converge based on the social power vector $c^\top$.

$$c^\top x(k+1) = c^\top x(k) + c^\top u(k)$$

2. The Final State

For $c^\top x(T)$, the timing of the input makes no difference.



$$c^\top x(T) = c^\top x(0) + \sum_{k=0}^{T-1} c^\top u(k)$$

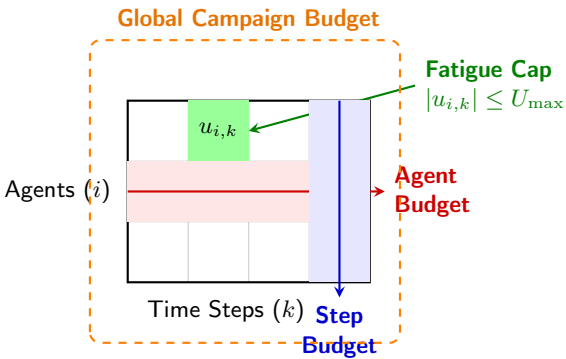
3. Objective

To maximize the effect of the input, we minimize the sum across all T steps:

$$\min_U \sum_{m=1}^{T} c^\top x(m)$$

Problem Formulation: Constraints

Constraints



Mathematical Formulation

Financial cost $\text{Cost}_{i,k} = p_i|u_{i,k}|$

1. **Global Budget** $\sum_k \sum_i \text{Cost}_{i,k} \leq B_{\max}$
2. **Step Budget** $\sum_i \text{Cost}_{i,k} \leq B_{\text{step}} \quad \forall k$
3. **Agent Budget** $\sum_k \text{Cost}_{i,k} \leq B_{\text{Agent}} \quad \forall i$
4. **Fatigue Cap** $|u_{i,k}| \leq U_{\max} \quad \forall i, k$

Complete Optimization Problem

$$
\begin{aligned}
\min_{U,X} \quad & \sum_{m=1}^{T} c^\top x(m) \\
\text{subject to} \quad & x(k+1) = Wx(k) + u(k), && \forall k \in \{0, \dots, T-1\} \\
& x(0) = x_{\text{initial}} \\
& \sum_{k=0}^{T-1} \sum_{i=1}^{n} p_i |u_{i,k}| \leq B_{\max} && \text{(Global Budget)} \\
& \sum_{i=1}^{n} p_i |u_{i,k}| \leq B_{\text{step}}, && \forall k \in \{0, \dots, T-1\} \\
& \sum_{k=0}^{T-1} p_i |u_{i,k}| \leq B_{\text{Agent}}, && \forall i \in \{1, \dots, n\} \\
& |u_{i,k}| \leq U_{\max}, && \forall i \in \{1, \dots, n\}, \forall k \in \{0, \dots, T-1\}
\end{aligned}
$$

Simulation Setup

Objective: Maximize downward shift of opinions over $T = 6$ steps ($n = 5$ agents).

1. Constraints

Global Budget: $\sum_{i,j} Cost_{i,j} \leq 15.0$

Step Budget: $\sum_i Cost_{i,k} \leq 3.0$

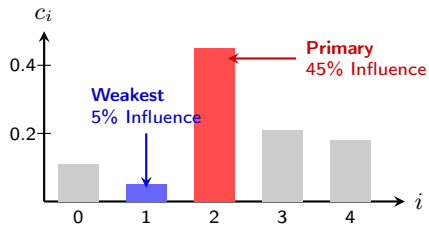
Person Budget: $\sum_i Cost_{i,k} \leq 3.0$

Fatigue Cap: $|u_{i,k}| \leq 1.0$

Financial Cost:

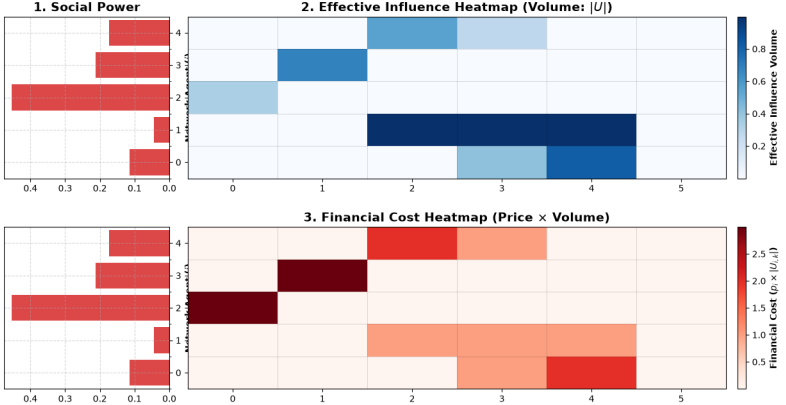
$$\text{Cost}_{i,k} = p_i|u_{i,k}|,\ p_i = \left(c_i/\min(c)\right)^{\gamma}$$

2. Social Power (c_i)



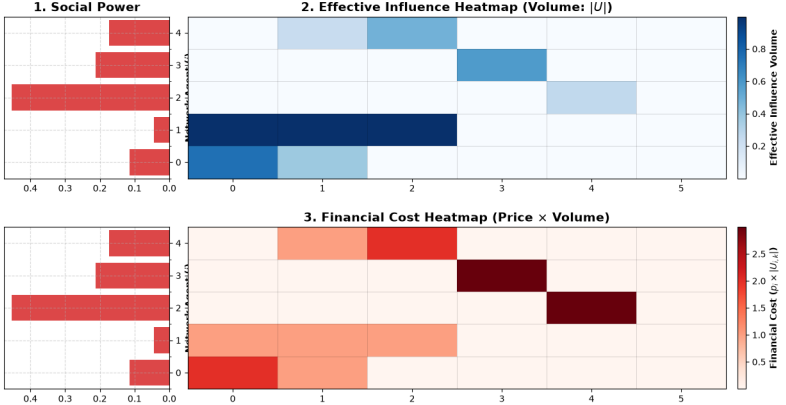
Simulation Results: $\gamma = 0.95$ (Celebrity Bargain)

When $\gamma < 1$, top influencers are underpriced relative to their social power.



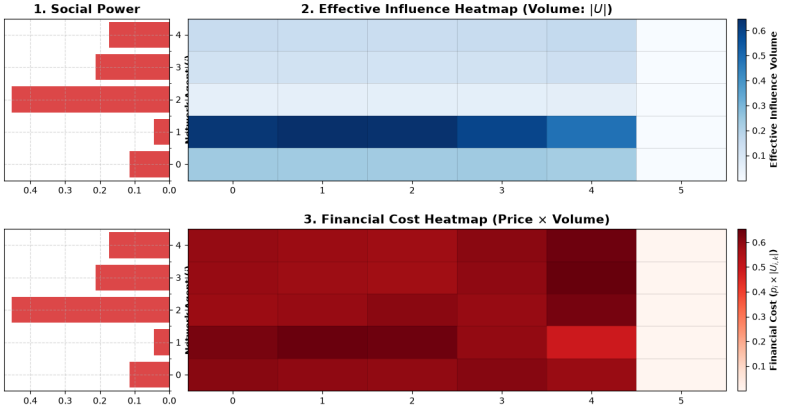
Simulation Results: $\gamma = 1.05$ (Celebrity Premium)

When $\gamma > 1$, a "celebrity premium" makes influential agents vastly overpriced.



Simulation Results: $\gamma = 1.0$ (Linear)

When $\gamma = 1.0$, every single agent in the network has the exact same cost-efficiency.



Thank You!

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